

TASK REPORT

Submitted to

MyCaptain

Course Name: Data Analytics With Python

Task no.1 : **AI Business Question & Data Quality Sprint**

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Table of Contents

Section	Page No.
Introduction to the Dataset	2
Business Questions	3
Initial Data Quality Findings	4
AI-Supported Cleanup Plan	7
Next-Step Analysis Plan	7
Reflection	8

GitHub link: [AI Business Question & Data Quality Sprint](#)

SECTION 1 — DATASET OVERVIEW

id	show_id	type	title	director	cast	country	date_added/release_date	rating	duration	listed_in	description
1	1	Movie	Doc John	Kristen Johnson	United States	September 2020	PG-13	90 min	Documentary	As her father nears the end of his life, filmmaker Kristen Johnson stages his death in inventive and comical ways to help them both face the inevitable.	
2	2	TV Show	Shoe & Winter	Anna Quins	South Africa	September 2021	TV-MA	2 Seasons	Intimate	After crossing paths at a party, a Cape Town teen sets out to prove whether a private-school scholastic star is her sister who was abducted at birth.	
3	3	TV Show	Ganglands	Lilian Luc	Saint-Sauveur, Haiti	September 2021	TV-MA	1 Season	Crime TV	To protect his family from a powerful drug lord, skilled thief Mahat and his expert team of robbers are pulled into a violent and deadly turf war.	
4	4	TV Show	Julien's New Orleans			September 2021	TV-MA	1 Season	Documentary	Julien, a filmmaker and writer, tells his story among the incarcerated women at the Orleans Justice Center in New Orleans on this gritty reality series.	
5	5	TV Show	Kaka Factory	Blaise Mounier		September 2021	TV-MA	2 Seasons	Intimate	In a city of swirling rumors known to their friends as "the island," the first college-level student and his friends navigate campus life.	
6	6	TV Show	Madrigal: The Movie	Kate Siegel, Jacki Koff		September 2021	TV-MA	1 Season	TV Drama	The arrival of a charismatic young priest brings glorious miracles, personal mystery and renewed religious fervor to a dying town desperate to believe.	
7	7	Movie	My Little #Babe	Cul Valerius Hughes		September 2021	PG	95 min	Children & Family	A divided, but a bright-eyed hero believes Earth's Prince, Paget and his team should be paid. And, heart to heart, should be determined to prove it.	
8	8	Movie	Sankofa	Halle Geri Hall	Ghana	United States	September 1993	TV-MA	125 min	Drama	In On a photo shoot in Ghana, an American model slips back in time, becomes enslaved on a plantation and bears witness to the agony of her ancestral past.
9	9	TV Show	The Great	Andy Davis	Malta	United States	September 2021	TV-14	8 Seasons	British TV	A selected batch of amateur bakers face off in a 10-week competition, whipping up their best dishes in the hopes of being named the U.K.'s best.
10	10	Movie	The Marine	Therese Maltre	United States	September 2021	PG-13	104 min	Comedy	A woman adjusting to life after a loss contends with a fussy bird that's taken over her garden bed and a husband who's struggling to find a way forward.	
11	11	TV Show	Indelible Truth	Lies and the Mafia		September 2021	TV-MA	1 Season	Crime TV	Italy's last mafia boss is a hard "Yakuza" coalition. But what happens when those trying to bring them organized crime are accused of being criminals themselves?	
12	12	TV Show	Bangkok Bangkok	K Subhachul	Thailand	September 2021	TV-MA	1 Season	Crime TV	Struggling to earn a living in Bangkok, a man joins an emergency rescue service and realizes he must unravel a citywide conspiracy.	
13	13	Movie	Je Suis Kar Christian	Luca Wedel	Germany	September 2021	TV-MA	127 min	Drama	In the heart of the family in a terrorist bombing, a young woman is unconsciously turned into joining the very group that killed them.	
14	14	Movie	Confession	Bruno Garcia	Spain	September 2021	TV-MA	95 min	Children & Family	When the clever but socially awkward Tessa joins a new school, she'll do anything to fit in. But the queen bee among her classmates has other ideas.	
15	15	TV Show	Crime Stories	India	Detectives	September 2021	TV-MA	1 Season	British TV	Cameras following Bengali police on the job offer a rare glimpse into the complex and challenging inner workings of four major crime investigations.	
16	16	TV Show	Dear White People	Logan Pines	United States	September 2021	TV-MA	4 Seasons	TV Comedy	Students of color navigate the daily slights and slippery politics of life at an Ivy League college that's not nearly as "post-racial" as it thinks.	
17	17	Movie	Escapade	Alfred de Lencastre	Spain	September 2020	TV-MA	97 min	Documentary	Declassified documents reveal the post-WWII life of Otto Skorzeny, a close Hitler ally who escaped to Spain and became an advisor to world presidents.	

About the Dataset

The dataset contains information about Movies and TV Shows available on Netflix. It includes attributes such as content type, title, director, cast, country, release year, rating, duration, genre, and description.

The dataset helps analyze Netflix's content library and supports business understanding of content trends, audience targeting, and regional entertainment patterns.

Stakeholders Who May Use This Dataset

Business Analysts

- Analyze genre trends
- Study regional demand
- Compare Movies vs TV Shows

Netflix Management

- Plan content investment
- Improve subscriber engagement
- Support strategic expansion

Recommendation Teams

- Personalize user recommendations
- Improve watch-time and retention

Marketing Teams

- Design targeted promotional campaigns
- Promote seasonal or trending genres

Data Scientists

- Build recommendation systems
- Perform trend forecasting and predictive analysis

Business Decisions Supported

The dataset can support decisions related to:

- Content investment planning
- Regional expansion strategy
- Audience targeting
- Genre performance analysis
- Release scheduling
- Customer retention strategies

Example insights include:

- identifying fast-growing genres,
 - understanding country-wise content trends,
 - and analyzing whether Movies or TV Shows dominate the platform.
-

Overall Business Value

This dataset provides business value by helping streaming platforms:

- improve recommendations,
 - optimize content investments,
 - increase subscriber engagement,
 - and support data-driven decision making in the OTT entertainment industry.
-

How AI Was Used

AI was used to help organize dataset understanding, identify business use cases, and structure stakeholder analysis.

SECTION 2 — BUSINESS QUESTIONS

Business Question 1

Which genre contains the highest number of Movies and TV Shows on Netflix?

This question helps identify the most popular content categories on Netflix and supports decisions related to content investment and audience preferences.

Business Question 2

Which countries produce the most Netflix content?

This question helps understand regional content distribution and supports business decisions related to global expansion and localized content production.

Business Question 3

What type of content (Movies or TV Shows) is added more frequently in recent years?

This question helps analyze Netflix's changing content strategy and understand whether the platform is focusing more on Movies or long-form TV Shows.

SECTION 3 — CHECK DATA QUALITY

STEP 1 — Load Dataset

```
import pandas as pd

df = pd.read_csv("/content/netflix_titles.csv")
```

Top 5 rows using head()

```
print(df.head())
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-13	90 min
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mablane, Thaban...	South Africa	September 24, 2021	2021	TV-MA	2 Seasons
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021	TV-MA	1 Season
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021	TV-MA	1 Season
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021	TV-MA	2 Seasons

```

                                listed in \
0                                Documentaries
1  International TV Shows, TV Dramas, TV Mysteries
2  Crime TV Shows, International TV Shows, TV Act...
3                                Docuseries, Reality TV
4  International TV Shows, Romantic TV Shows, TV ...

                                description
0  As her father nears the end of his life, filmm...
1  After crossing paths at a party, a Cape Town t...
2  To protect his family from a powerful drug lor...
3  Feuds, flirtations and toilet talk go down amo...
4  In a city of coaching centers known to train I...

```

Dataset Structure and Initial Inspection

STEP 2 — Check Dataset Info

```

print(df.info())

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
 #   Column          Non-Null Count  Dtype
---  ---
 0   show_id         8807 non-null   object
 1   type            8807 non-null   object
 2   title           8807 non-null   object
 3   director        6173 non-null   object
 4   cast            7982 non-null   object
 5   country         7976 non-null   object
 6   date_added      8797 non-null   object
 7   release_year    8807 non-null   int64
 8   rating          8803 non-null   object
 9   duration        8804 non-null   object
10   listed_in       8807 non-null   object
11   description     8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
None

```

The dataset contains 8,807 records and 12 columns related to Netflix Movies and TV Shows.

- Most columns use the object data type because they contain text-based information such as title, director, country, and genre.
- The release_year column uses the integer data type since it contains numerical year values.

STEP 3 — Check Missing Values

```

print(df.isnull().sum())

... show_id         0
    type            0
    title           0
    director       2634
    cast           825
    country        831
    date_added      10
    release_year     0
    rating          4
    duration        3
    listed_in       0
    description     0
    dtype: int64

```

Initial Findings

- The director, cast, and country columns contain a significant number of missing values and may require cleaning before analysis.
- Missing values in date_added, rating, and duration should also be reviewed to avoid incomplete analysis results.
- The dataset is mostly structured and suitable for business analysis after preprocessing and data cleaning.

STEP 4 — Check Duplicate Rows

```
print(df.duplicated().sum())
```

No duplicate values

STEP 5 — Check Unique Values

```
print(f"Types: {df['type'].unique()}")
print(f"Ratings: {df['rating'].unique()}")

Types: ['Movie' 'TV Show']
Ratings: ['PG-13' 'TV-MA' 'PG' 'TV-14' 'TV-PG' 'TV-Y' 'TV-Y7' 'R' 'TV-G' 'G'
'NC-17' '74 min' '84 min' '66 min' 'NR' nan 'TV-Y7-FV' 'UR']
```

The type column contains only two valid categories:

- Movie
- TV Show

However, the rating column contains unusual values such as:

- 74 min
- 84 min
- 66 min

These values appear to be duration values placed incorrectly in the rating column, indicating possible formatting or data entry issues.

STEP 6 — Check Outliers / Suspicious Values

```
print(df["release_year"].describe())
```

count	8807.000000
mean	2014.180198
std	8.819312
min	1925.000000
25%	2013.000000
50%	2017.000000
75%	2019.000000
max	2021.000000
Name: release_year, dtype: float64	

Release Year Inspection

The release_year column contains values ranging from **1925 to 2021**.

Most content appears to be released between **2013 and 2019**, indicating that Netflix mainly contains modern content.

The older values should be reviewed manually to ensure they are valid and not data entry errors.

SECTION 4 — AI-Supported Cleanup Plan

AI-Supported Cleanup Plan

Issue Identified	Cleanup Action	Why It Matters
Missing values in director, cast, and country	Replace with “Unknown” or review manually	Prevent incomplete analysis
Incorrect values in rating column	Correct or remove misplaced duration values	Ensure accurate rating analysis
Missing values in date_added, rating, and duration	Review and fill where possible	Avoid incomplete records
Possible duplicate records	Remove duplicate rows	Prevent inaccurate content counts
Mixed duration formats	Separate Movies and TV Shows before analysis	Improve duration analysis accuracy

How AI Was Used

AI was used to suggest possible cleanup steps and identify common quality issues in categorical data.

SECTION 5 — Next-Step Analysis Plan

Next-Step Analysis Plan

After cleaning the dataset, the following analyses can be performed:

- Genre popularity analysis
- Country-wise content distribution analysis
- Movie vs TV Show trend analysis
- Content rating distribution analysis
- Yearly content growth analysis

These analyses can help understand Netflix’s content strategy and audience preferences.

How AI Was Used

AI was used to suggest possible business analysis directions based on the dataset structure.

SECTION 6 — Reflection

AI helped speed up the process of identifying possible data quality issues, cleanup strategies, and business-focused questions.

However, human judgment was required to decide which issues were important and which records needed manual verification.

Before presenting final findings, I would manually review:

- missing country information,
- unusual rating values,
- and possible duplicate records.

How AI Was Used

AI was used as a support tool for planning and organizing the analysis process, while final decisions were made manually.